Compression Strategies for High-Throughput X-Ray CT of Natural History Collections

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The NaturArv3D project[1] will acquire 3D X-ray images of the full vertebrate collection of the Natural History Museum Denmark at sub-10 µm voxel size, in collaboration with the DTU 3D Imaging Centre. The scale of the project necessitates a custom high-throughput X-ray scanner with robotic sample handling, capable of scanning up to 500 samples per week. The expected annual data volume of approximately 1 PB renders long-term archival storage economically onerous without substantial compression.

The compression ratios (CR), compression and decompression times, and quality loss are determined for in-memory compression tests across many lossless and lossy compression schemes, including JPEG-LS and the recent JPEG XL standard. Relevant subsets of the tests are performed in TIFF, DICOM, and Zarr containers in order to determine the most viable archival formats and their compatibility with prevalent image-processing software packages. Quality loss is quantified using common full-reference metrics including SSIM [1], MS-SSIM [2], and PSNR.

We find that lossless schemes yield a maximum CR of 1.75:1, corresponding to a 43% space saving (SS). The JPEG-LS and JPEG XL codecs achieve CRs of 16.3:1 (93.9% SS) and 45.6:1 (97.8% SS), respectively, at imperceptible quality loss ($\mathrm{PSNR} \approx 60\,\mathrm{dB}$ full scale, $3\mathrm{D\ SSIM} \approx 0.999$) on a representative 16-bit CT volume. Alongside JPEG 2000, these schemes in combination with a DICOM container constitute the only lossy options widely supported among the surveyed packages. Compression times remain compatible with high-throughput acquisition, while decompression is sufficiently fast to improve access times over the uncompressed, I/O-bound case.

Relative to the baseline data pipeline without compression, the proposed schemes offer an estimated annual storage-cost saving of 2 million DKK under realistic operating conditions. This underscores that the adoption of lossy compression is decisive for the economic viability of high-throughput X-ray imaging at scale.

References

[1] Z. Wang, A. Bovik, H. Sheikh, and E. Simoncelli, "Image Quality Assessment: From Error Visibility to Structural Similarity," *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600–612, Apr. 2004, doi: 10.1109/TIP.2003.819861.

[2] Z. Wang, E. Simoncelli, and A. Bovik, "Multiscale Structural Similarity for Image Quality Assessment," in *The Thrity-Seventh Asilomar Conference on Signals, Systems & Computers, 2003*, Nov. 2003, p. 1398–1402 Vol.2. doi: 10.1109/ACSSC.2003.1292216.

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[1]https://3dim.dtu.dk/newsarchive/2025/03/naturarv_3d